EX N0:3 07.02.2025	JOIN OPERATION

AIM:

To execute various types of Join.

JOINS:

Joins in SQL are commands which are used to combine rows from twoor more tables,based on a related column between those tables.

TYPES:

- Cross Join (or) Cartesian Product
- Inner join
- Natural Join
- Outer Join
    - Left Outer Join
    - Right Outer Join
    - Full Outer Join
- Self Join
- Equi Join
- Theta join

TABLE CREATION:

```
SQL> create table students(sid number(3), sname varchar2(10), cid number(3),
primary key(sid));

Table created.

SQL> create table courses(cid number(3), cname varchar2(10));

Table created.


SQL> select * from students;


       SID SNAME             CID
---------- ---------- ----------
         1 Amit              101
         2 Sita              102
         3 Rahul             104
```

```
         4 Priya              106
         5 Yokarajan          103

SQL> select * from courses;

       CID CNAME
---------- ----------
       101 Maths
       103 DBMS
       107 WT
       105 OS
       104 DAA
```

CROSS JOIN (OR) CARTESIAN PRODUCT :

CROSS JOINs are used to combine each row of one table with each row of another table, and return the Cartesian product of the sets of rows from the tables.

```
SQL> select * from students, courses;
       SID SNAME             CID        CID CNAME
---------- ---------- ---------- ---------- ----------
         1 Amit              101        101 Maths
         1 Amit              101        103 DBMS
         1 Amit              101        107 WT
         1 Amit              101        105 OS
         1 Amit              101        104 DAA
         2 Sita              102        101 Maths
         2 Sita              102        103 DBMS
         2 Sita              102        107 WT
         2 Sita              102        105 OS
         2 Sita              102        104 DAA
         3 Rahul             104        101 Maths
         3 Rahul             104        103 DBMS
         3 Rahul             104        107 WT
         3 Rahul             104        105 OS
         3 Rahul             104        104 DAA
```

```
        4 Priya              106        101 Maths
        4 Priya              106        103 DBMS
        4 Priya              106        107 WT
        4 Priya              106        105 OS
        4 Priya              106        104 DAA
        5 Yokarajan          103        101 Maths
        5 Yokarajan          103        103 DBMS
        5 Yokarajan          103        107 WT
        5 Yokarajan          103        105 OS
        5 Yokarajan          103        104 DAA

25 rows selected.
```

NATURAL JOIN(USING KEYWORD):

- A Natural join is the join operation where the common attributes occur only one time.

```
SQL> select * from students natural join courses;

       CID        SID SNAME      CNAME
---------- ---------- ---------- ----------
       101          1 Amit       Maths
       103          5 Yokarajan  DBMS
       104          3 Rahul      DAA
```

INNER JOIN:

Inner Join is type of join in the which the common attribute in the resultant table appear twice .The resultant table only contains records that have common attribute in both the table.

USING INNER JOIN KEYWORD:

```
SQL> select * from students s inner join courses c on s.cid = c.cid;
       SID SNAME                CID        CID CNAME
---------- ---------- ---------- ---------- ----------
         1 Amit                 101        101 Maths
```

```
         5 Yokarajan          103         103 DBMS
         3 Rahul              104         104 DAA
```

INNER JOIN WITH ("USING" CLAUSE):

```
SQL> select * from students inner join courses using(cid);

       CID        SID SNAME      CNAME
---------- ---------- ---------- ----------
       101          1 Amit       Maths
       103          5 Yokarajan  DBMS
       104          3 Rahul      DAA
```

WITHOUT USING KEYWORD:

```
SQL> select * from students s, courses c where s.cid = c.cid;
       SID SNAME             CID        CID CNAME
---------- ---------- ---------- ---------- ----------
         1 Amit              101        101 Maths
         5 Yokarajan         103        103 DBMS
         3 Rahul             104        104 DAA
```

OUTER JOIN:

An outer join is a method of combining two or more tables so that the result includes unmatched rows of one of the tables, or of both tables.

LEFT OUTER JOIN(USING KEYWORD):

A left outer join in SQL merges all records from the left table with matching records from the right table based on a given condition, including unmatched records from the left table and filling in NULL values for missing matches from the right table.

```
SQL> select * from students s left outer join courses c on s.cid=c.cid;

       SID SNAME             CID        CID CNAME
---------- ---------- ---------- ---------- ----------
         1 Amit              101        101 Maths
         5 Yokarajan         103        103 DBMS
```

```
         3 Rahul              104        104 DAA
         4 Priya              106
         2 Sita               102
```

LEFT OUTER JOIN(USING + OPERATOR):

```
SQL> select * from students s, courses c where s.cid=c.cid(+);

       SID SNAME              CID        CID CNAME
---------- ---------- ---------- ---------- ----------
         1 Amit               101        101 Maths
         5 Yokarajan          103        103 DBMS
         3 Rahul              104        104 DAA
         4 Priya              106
         2 Sita               102
```

RIGHT OUTER JOIN:

A right outer join in SQL combines all rows from the right table with matching rows from the left table based on a specified condition, including unmatched rows from the right table and filling in NULL values for columns from the left table where no match is found.

RIGHT OUTER JOIN(USING KEYWORD):

```
SQL> select * from students s right outer join courses c on s.cid=c.cid;

       SID SNAME              CID        CID CNAME
---------- ---------- ---------- ---------- ----------
         1 Amit               101        101 Maths
         3 Rahul              104        104 DAA
         5 Yokarajan          103        103 DBMS
                                         105 OS
                                         107 WT
```

RIGHT OUTER JOIN(USING + OPERATOR):

```
SQL> select * from students s, courses c where s.cid(+)=c.cid
```

```
       SID SNAME             CID        CID CNAME
---------- ---------- ---------- ---------- ----------
         1 Amit              101        101 Maths
         3 Rahul             104        104 DAA
         5 Yokarajan         103        103 DBMS
                                        105 OS
                                        107 WT
```

FULL OUTER JOIN:

A FULL OUTER JOIN in SQL merges all rows from both tables, including unmatched rows from both tables, based on a specified condition, filling in NULL values for columns where no match is found in either table.

FULL OUTER JOIN(USING KEYWORD):

```
SQL> select * from students s full outer join courses c on s.cid=c.cid;

       SID SNAME             CID        CID CNAME
---------- ---------- ---------- ---------- ----------
         1 Amit              101        101 Maths
         5 Yokarajan         103        103 DBMS
         3 Rahul             104        104 DAA
         4 Priya             106
         2 Sita              102
                                        105 OS
                                        107 WT

7 rows selected.
```

EQUI JOIN:

An EQUI JOIN in SQL merges rows from two tables based on matching values in specified columns, using the equality operator in the join condition.

```
SQL> select c.cname from students s, courses c where s.cid=c.cid;
```

```
CNAME
----------
Maths
DBMS
DAA
```

THETA JOIN:

Theta join is a type of join condition where the condition involves a comparison operator other than just "=" operator.

```
SQL> select * from stud3;

       LID LNAME           MARKS
---------- ---------- ----------
         1 Aarav              95
         2 Aisha              88
         3 Rohan              72
         4 Yokarajan          85
         5 Vihaan             78

SQL> select * from subj3;
       LID SUBNAME    GR
---------- ---------- --
         1 Math       A
         2 DBMS       O
         3 OS         A+
         4 WT         B+
         5 DAA        A
```

USING ("=" OPERATOR):

```
SQL> select s.lname, s.marks from stud3 s, subj3 sub where s.lid = sub.lid and
s.marks > 80;
```

```
LNAME           MARKS
---------- ----------
Aarav              95
Aisha              88
Yokarajan          85
```

USING (!= OPERATOR):

```
SQL> select * from stud3 s, subj3 sub where s.lid != sub.lid;


       LID LNAME           MARKS        LID SUBNAME    GR
---------- ---------- ---------- ---------- ---------- --
         1 Aarav              95          2 DBMS       O
         1 Aarav              95          3 OS         A+
         1 Aarav              95          4 WT         B+
         1 Aarav              95          5 DAA        A
         2 Aisha              88          1 Math       A
         2 Aisha              88          3 OS         A+
         2 Aisha              88          4 WT         B+
         2 Aisha              88          5 DAA        A
         3 Rohan              72          1 Math       A
         3 Rohan              72          2 DBMS       O
         3 Rohan              72          4 WT         B+
         3 Rohan              72          5 DAA        A
         4 Yokarajan          85          1 Math       A
         4 Yokarajan          85          2 DBMS       O
         4 Yokarajan          85          3 OS         A+
         4 Yokarajan          85          5 DAA        A
         5 Vihaan             78          1 Math       A
         5 Vihaan             78          2 DBMS       O
         5 Vihaan             78          3 OS         A+
         5 Vihaan             78          4 WT         B+
20 rows selected.
```

USING (> OPERATOR):

```
SQL> select * from stud3 s, subj3 sub where s.lid > sub.lid;
```

```
       LID LNAME           MARKS        LID SUBNAME    GR
---------- ---------- ---------- ---------- ---------- --
         2 Aisha              88          1 Math       A
         3 Rohan              72          1 Math       A
         3 Rohan              72          2 DBMS       O
         4 Yokarajan          85          1 Math       A
         4 Yokarajan          85          2 DBMS       O
         4 Yokarajan          85          3 OS         A+
         5 Vihaan             78          1 Math       A
         5 Vihaan             78          2 DBMS       O
         5 Vihaan             78          3 OS         A+
         5 Vihaan             78          4 WT         B+
10 rows selected.
```

USING (>=OPERATOR):

```
SQL> select * from stud3 S, subj3 SJ where S.lid >= SJ.lid;

      LID  LNAME          MARKS       LID   SUBNAME    GR
---------- ---------- ---------- ---------- ---------- --
         1 Aarav              95          1 Math       A
         2 Aisha              88          1 Math       A
         2 Aisha              88          2 DBMS       O
         3 Rohan              72          1 Math       A
         3 Rohan              72          2 DBMS       O
         3 Rohan              72          3 OS         A+
         4 Yokarajan          85          1 Math       A
         4 Yokarajan          85          2 DBMS       O
         4 Yokarajan          85          3 OS         A+
         4 Yokarajan          85          4 WT         B+
         5 Vihaan             78          1 Math       A
         5 Vihaan             78          2 DBMS       O
         5 Vihaan             78          3 OS         A+
         5 Vihaan             78          4 WT         B+
         5 Vihaan             78          5 DAA        A
15 rows selected.
```

USING (<=OPERATOR):

```
SQL> select * from stud3 S, subj3 SJ where S.lid <= SJ.lid;

      LID  LNAME           MARKS       LID SUBNAME     GR
---------- ---------- ---------- ---------- ---------- --
         1 Aarav              95          1 Math       A
         1 Aarav              95          2 DBMS       O
         1 Aarav              95          3 OS         A+
         1 Aarav              95          4 WT         B+
         1 Aarav              95          5 DAA        A
         2 Aisha              88          2 DBMS       O
         2 Aisha              88          3 OS         A+
         2 Aisha              88          4 WT         B+
         2 Aisha              88          5 DAA        A
         3 Rohan              72          3 OS         A+
         3 Rohan              72          4 WT         B+
         3 Rohan              72          5 DAA        A
         4 Yokarajan          85          4 WT         B+
         4 Yokarajan          85          5 DAA        A
         5 Vihaan             78          5 DAA        A
15 rows selected.
```

USING (<OPERATOR):

```
SQL>  select * from stud3 S, subj3 SJ where S.lid < SJ.lid;

       LID LNAME           MARKS        LID SUBNAME    GR
---------- ---------- ---------- ---------- ---------- --
         1 Aarav              95          2 DBMS       O
         1 Aarav              95          3 OS         A+
         1 Aarav              95          4 WT         B+
         1 Aarav              95          5 DAA        A
         2 Aisha              88          3 OS         A+
         2 Aisha              88          4 WT         B+
         2 Aisha              88          5 DAA        A
         3 Rohan              72          4 WT         B+
```

```
         3 Rohan              72          5 DAA        A
         4 Yokarajan          85          5 DAA        A
10 rows selected.
```

SELF JOIN:

A SELF JOIN in SQL is used to join a table to itself, typically to compare rows within the same table by assigning different aliases to the same table in the query.

```
SQL> select * from stucou;

    STU_ID SNAME          COU_ID
---------- ---------- ----------
       101 Arjun
       102 Bhuvi             107
       103 Yokarajan         101
       104 Deepak            101
       105 Raju              102
       106 Ram               102

6 rows selected.

SQL> select s1.sname, s2.sname from stucou s1, stucou s2 where s1.stu_id =
s2.cou_id;

SNAME      SNAME
---------- ----------
Arjun      Yokarajan
Arjun      Deepak
Bhuvi      Raju
Bhuvi      Ram
```

JOINING THREE TABLES:

```
SQL> select * from students;

       SID SNAME             CID
---------- ---------- ----------
         1 Amit              101
         2 Sita              102
         3 Rahul             104
```

```
         4 Priya              106
         5 Yokarajan          103
```

```
SQL> select * from courses;
       CID CNAME
---------- ----------
       101 Maths
       103 DBMS
       107 WT
       105 OS
       104 DAA
```

```
SQL> select * from stud_marks;
       LID        CID SUBNAME              GR
---------- ---------- -------------------- --
         1        101 Math                 A
         2        103 DBMS                 O
         3        105 OS                   A+
         4        107 WT                   B+
         5        104 DAA                  A
```

JOIN USING THREE TABLES:

```
SQL> select s.sname, c.cname, m.gr from students s, courses c, stud_marks m where
s.cid = c.cid and s.cid=m.cid and c.cid = m.cid;

SNAME      CNAME      GR
---------- ---------- --
Amit       Maths      A
Yokarajan  DBMS       O
Rahul      DAA        A
```

CONTENTS	MARKS ALLOTED	MARKS OBTAINED
Aim, Algorithm, SQL, PL/SQL	30	
Execution and Result	20	
Viva	10	
Total	60	

RESULT:

Thus ,various types of join commands are executed sucessfully.